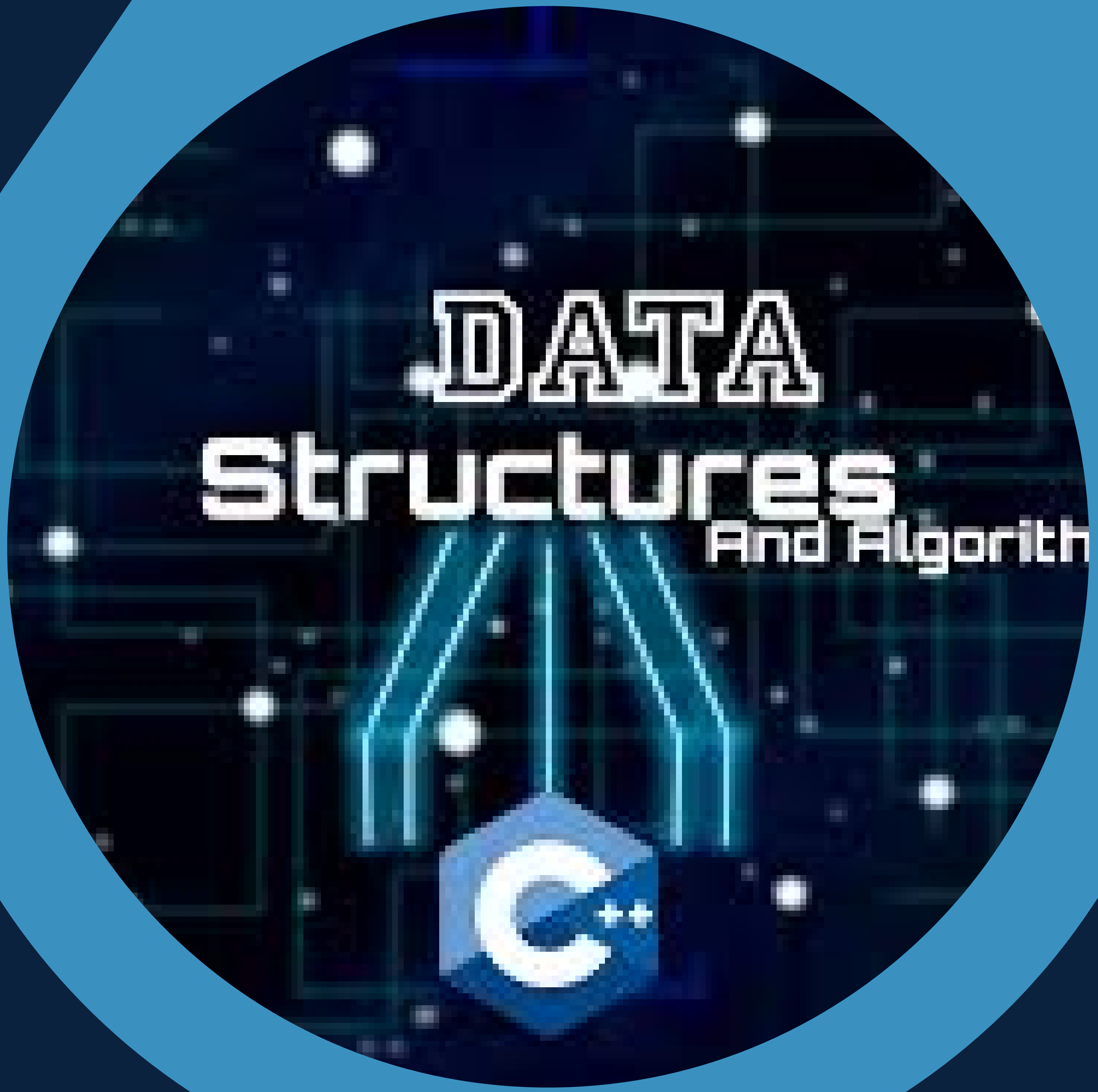
DSA Foundation Program



Presented by Livesh Garg

Day 1: Introduction to DSA

- What is DSA?
- Why companies ask DSA?
- Problem-solving process
- Flowchart thinking

Practice:

- Print patterns
- Basic logic questions

Day 2: Time & Space Complexity

- O(1)
- O(n)
- O(log n)
- Best, Average, Worst Case

Practice:

Complexity identification

Day 3: Variables & Problem Breakdown

- How to understand a problem
- Input → Process → Output

Practice:

- Sum
- Average
- Maximum number

Day 4: Loops & Thinking

- for loop
- while loop
- Nested loops

Practice:

- Sum
- Average
- Maximum number

Day 5: Functions & Modular Thinking

- Functions
- Parameters
- Return values

Practice:

- Calculator
- Prime checker

Day 6: Arrays Basics

- Creation
- Traversal
- Access

Practice:

- Find largest element

Day 7: Array Searching

- Linear Search

Practice:

- Search element

LeetCode:

- Two Sum

Day 8: Array Manipulation

- Insert
- Delete
- Update

Practice:

- Remove duplicates

Day 9: Prefix Sum

- Range Sum Queries

Practice:

- Running Sum

LeetCode:

- Running Sum of 1D Array

Day 10: Sliding Window

- Fixed Window
- Variable Window

Practice:

- Maximum Sum Subarray

Day 11: String Basics

- Character access
- Length
- Traversal

Practice:

- Reverse String

Day 12: Frequency Counting

- Character count
- Hashing concept

Practice:

- Anagram Check

Day 13: String Patterns

- Palindrome
- Substrings

Practice:

- Valid Palindrome

Day 14: Recursion Basics

- Stack concept
- Base condition

Practice:

- Factorial

Day 15: Recursive Thinking

- Reverse Number
- Sum of digits

Day 16: Backtracking Introduction

- Choice making
- Decision tree

Practice:

- Generate subsets

Day 17: Hash Maps

- Key-Value concept

Practice:

- Frequency Count

Day 18: Hashing Patterns

LeetCode:

- Contains Duplicate
- Majority Element

Day 19: Linked List Basics

- Node
- Pointer

Practice:

- Traversal

Day 20: Insert & Delete

Practice:

- Insert Beginning
- Insert End

Day 21: Important Questions

LeetCode:

- Reverse Linked List
- Middle Node

Day 22: Stack

Practice:

- Balanced Parentheses

Day 23: Queue

Practice:

- Queue Implementation

Day 24: Monotonic Stack

- Insert Beginning
- Insert End

LeetCode:

- Next Greater Element

Day 25: Binary Tree Basics

- Root
- Child
- Height

Day 26: DFS Traversal

- Preorder
- Inorder
- Postorder

Day 27: BFS Traversal

LeetCode:

- Maximum Depth of Binary Tree

Day 28: Binary Search

- Search Space Thinking

LeetCode:

- Binary Search

Day 29: Greedy

- Local vs Global Optimum

Practice:

- Coin Change (Greedy Version)

Day 30: Revision & Mock Interview

- Arrays
- Strings
- Hashing
- Linked List
- Stack
- Queue
- Trees

Mock Coding Round

Daily 1-Hour Class Structure

First 10 Minutes
- Concept Explanation

Next 10 Minutes
- Dry Run on Whiteboard

Next 15 Minutes
- Teacher Solves One Problem

Next 15 Minutes
- Student Solves Similar Problem

Last 10 Minutes
- Discussion & Optimization

Problem Solving Framework (Most Important)

Step 1
- Understand Input

Step 2
- Understand Output

Step 3
- Write Example

Step 4
- Find Pattern

Step 5
- Write Brute Force

Step 6
- Optimize

Step 7
- Code

Step 8
- Dry Run

Step 9
- Analyze Complexity

Step 10
- Submit & Improve

Target After Completion

- Think logically before coding
- Understand LeetCode questions independently
- Solve Easy problems confidently
- Solve 50+ DSA questions
- Understand interview patterns
- Start Intermediate DSA (Graphs, Dynamic Programming, Tries, Advanced Trees) confidently

Contact Us

FOR FURTHER INQUIRIES, FEEL FREE TO REACH OUT.

Thank You!

WE APPRECIATE YOUR TIME.